

MICHAEL BICKFORD

Software Developer — Tools, Systems & Interactive Software

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Software developer and independent researcher focused on developer tooling, diagnostic systems, and interactive applications. I move between implementation and theory, using code, mathematical models, and structured investigation to make complex systems easier to understand, test, and use.

SELECTED WORK

What Do You Mean? — Live Diagnostic Environment

Private Capstone · 2026

Java 21 · Minecraft Fabric 1.21

Private capstone for diagnosing a live modded Minecraft client. Combines inspection, health monitoring, source access, and controlled debugging with explicit approval and rollback.

- Designed a staged diagnostic workflow: read-only inspection first, explicit approval for state changes, and rollback on failure.
- Implemented source, registry, and class inspection alongside health sampling and live condition monitoring.
- Documented outcomes and limitations through post-mortems; design documentation and a recorded demo are available on request.

Manifold — Git Client for Windows & Linux

Live · Community Source

Flutter / Dart

Desktop Git client for exploring file relationships around a change and navigating unfamiliar codebases.

- Started as a Tauri/Rust app; rebuilt in Flutter/Dart when the original proved too slow. Currently working on graph exploration, structured diffs, and Git-native review flows.

Minecraft Server Maintainer

Open Source

Java 21 utility · 19 GitHub stars · 8 server platforms

Headless utility for Minecraft servers, mods, plugins, and datapacks across 8 platforms. Provides pre-update checks, automated rollback on failure, and repeatable maintenance workflows.

RESEARCH & WRITING

The Self-Referential Fixed Point of the Complex Exponential

arXiv:2606.01668 · June 2026

Independent work on a fixed point of the complex exponential. Also writes technical documentation on live systems and high-dimensional mathematics.

CAPABILITIES & TOOLS

SOFTWARE DEVELOPMENT

Architecture, debugging, testing, review, documentation

INTERACTIVE SYSTEMS

Gameplay, tools, UX, live iteration

INVESTIGATION

Source reading, blame tracing, failure analysis

AUTOMATION

Tooling, workflows, integrations

LANGUAGES

C++ · Java · Python · Dart · Rust · TypeScript

APPLICATION & WEB

Flutter · HTML/CSS · Astro · WebAssembly integration

PLATFORMS

Fabric/Minecraft · Unreal Engine (C++) · Windows · Linux

PRACTICE

Git · testing · technical documentation

CAPSTONE: DIAGNOSTIC TOOLING FOR MODDED MINECRAFT

A private capstone built as a controlled diagnostic environment for a live modded client. The work combines inspection, health monitoring, source access, and user-approved state changes with rollback.

- **Live diagnosis.** Read-only inspection first, state changes only with explicit approval, rollback on failure.
- **Reactive conditions.** Compiled read-only triggers with safety checks, notifications, and background monitoring.
- **Source access.** File, jar, and registry scanning plus Vineflower decompilation, keeping code investigation close to the running environment.
- **Guarded execution.** Action-depth limits with visible confirmation before every state change.
- **Fidelity controls.** Privacy redaction that tells you exactly what was removed, with no silent gaps.
- **Evaluation.** Post-mortem separating measured results, observed behaviour, and open limitations.

GAME & INTERACTIVE WORK

Morithon

Team Project · 2024

Unreal Engine 5 death-run · 45-person student production

Built the main menu and settings UI, including anchored layouts, team-colour theming, and audio and graphics controls. Coordinated UI delivery with another core contributor when project ownership shifted.

Neuron

In Development

All-Rust Windows peripheral utility · raw HID, local configuration, automation

Rust utility for Razer peripherals, including direct HID interaction, Chroma SDK mock support, lighting controls, gesture input, and local TOML configuration.

EXPERIENCE

Open-Source & Independent Development

Ongoing

Building and documenting tools, prototypes, and studies across developer tooling, game modding, systems work, and math. Currently: WDYM, Manifold, Neuron, Minecraft Server Maintainer, and a solo arXiv paper.

Grounds Maintenance Worker — Humber College (Work-Study)

Summer 2023 – December 2025

Performed routine outdoor grounds maintenance and cleanup across campus as part of Humber's student work-study program, working reliably in a team through physical labour and changing weather conditions.

Customer Service — Tim Hortons

2017 – 2019 / 2021 – 2022

Drive-thru, beverage preparation, cash, and front-counter service in a fast-paced, high-volume environment.

EDUCATION

Advanced Diploma, Game Programming — Humber College

Currently Enrolled

Coursework in math, programming, leadership, and industry presence. Capstone: *What Do You Mean?*

Concurrent Education, Dramatic Arts

2019 – 2022

Three years completed, transferred during the pandemic. Psychology minor, improvisation, and community drama work.